Brijodison

Engineering Notebook Element J

Engineering Service Learning

Documentation of External Evaluation

We took in multiple accounts while conducting our prototyping including primary stakeholders as well as field experts. Our first input we received was from one of our field experts Jared Lawson, a graduate student researching engineering at Vanderbilt. He tried out our initial prototype and gave us valuable input that kickstarted us on our editing and redesign process. He said we could connect our two PVC poles in order to increase stability and make it easier to turn, and also added we could lose our poles if not bonded together by something. He also said when he can't get a tight enough fit he likes to put compressible materials in between two surfaces. Our second input was from a member of the sales force, Peter Hermann. He also told us we should bind the poles in order to increase stability, including the idea of crossing them like an "X". Seeing as they both gave a similar response, that was definitely the most urgent thing to address. We didn't want to add the handle due to attachment issues so we instead took one of Jared's ideas by adding rubber to the bottom point of contact. Rubber was the best material since it was easily compressible but still was able to create friction in order to prevent the sliding of our poles. It also acted as a binder between the surface of the PVC poles and any point of contact on the cart. We glued the rubber piece on the inside of the part where it attaches to the bottom of the cart.

Our third input we were given was from one of our primary stakeholders in Mr. Kiser. This device is first and foremost supposed to be compatible with him before anybody else. He began pushing the cart and quickly noticed that it slipped a lot due to there only being one point of contact with high friction (the part where the poles attach at the base), making it hard to turn. We realized that implementation of rubber along another point of contact would be optimal for the reduction of the poles sliding. We added a layer of rubber around our PVC pipe where it met the top part of the handle to combat this. Our fourth opinion was from Coach Sanford, another tall person who could benefit from our device. He said overall it was good and wasn't too hard to turn (our previous complaint), but still slid too much. We added a third point of contact using more rubber where the PVC pole met the bottom of the handle. With now having three points of contact, sliding isn't an issue and turning is now done comfortably.

Our final opinion was from Corbin Wolfgang, a student who uses this supply cart for robotics. His first complaint was that the PVC pipe was too bendy. This won't be a problem seeing as we will use non-malleable industrial metal on our final product. His other complaint was that the attachment to the cart slipped at the bottom. This won't happen on our final product because we are not going to cut the rubber tubing down the middle but rather keep it closed to ensure the bottom of the cart doesn't slide along the screwheads we used.

We took all feedback from our field experts as well as primary stakeholders in order to come to our best possible prototype. Not only did we adjust the prototype as we went along, but also used easily identified problems with the prototype to help envision our final product's materials and makeup. Outside opinion was one of the most important factors while constructing this prototype because it broadened the possibilities we would ever consider; not only identifying problems but also providing solutions we may have never thought of.